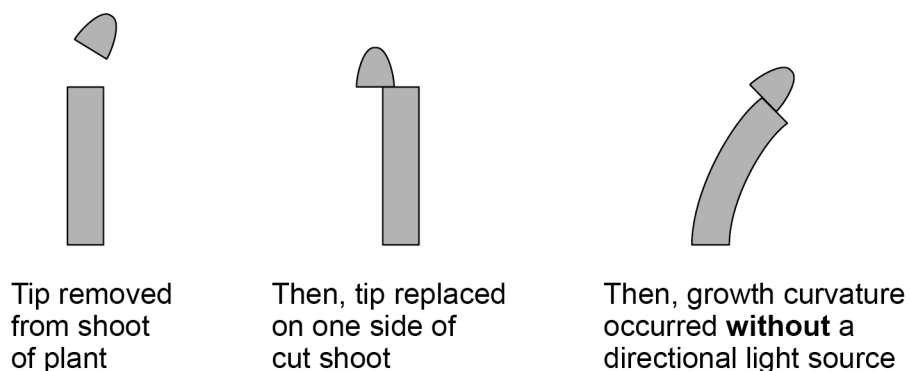


0 3

Figure 1 shows an investigation into growth factors in plants.

Figure 1



0 3

1

Use your knowledge of indoleacetic acid (IAA) to explain the growth curvature shown in **Figure 1**.

[3 marks]

Question 3 continues on the next page

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A bioassay is a method to determine the concentration of a substance by its effect on living tissues.

Figure 2 shows the practical procedure used in a growth curvature bioassay to determine the concentration of IAA in shoot tips.

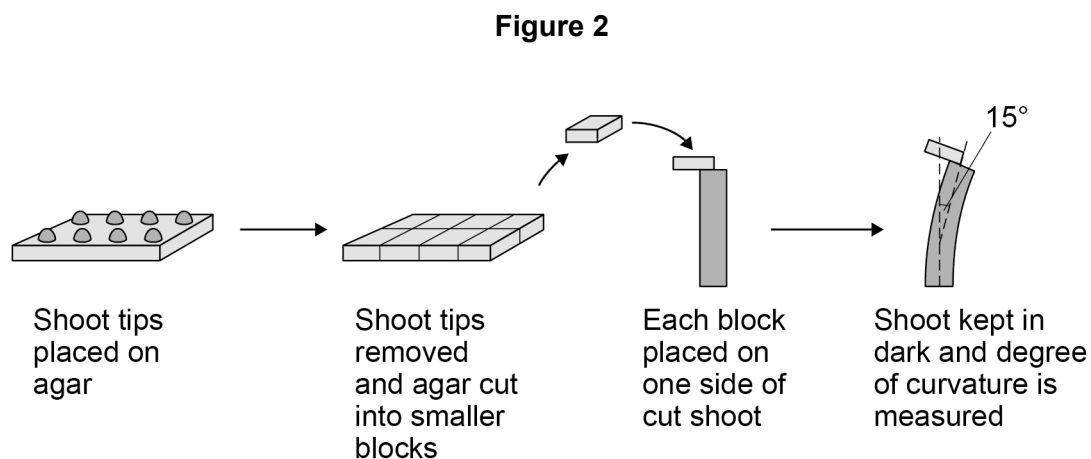
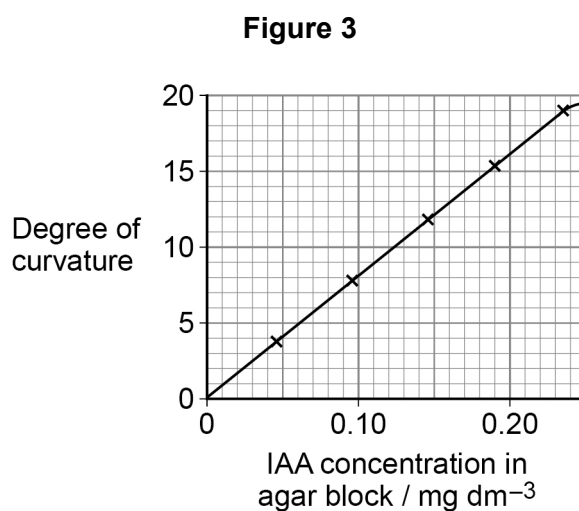


Figure 3 shows the calibration curve for this growth curvature bioassay.



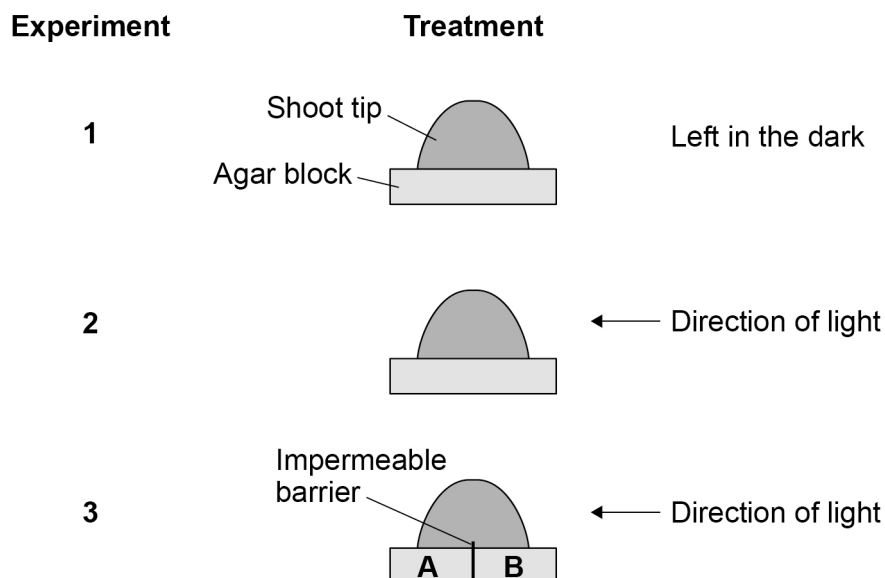
In your answer you should refer to all the variables that should be controlled to produce a valid comparison.

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A scientist investigated the effect of a directional light stimulus on the distribution of IAA in shoot tips. The scientist set up three experiments as shown in **Figure 4**. All variables were controlled apart from exposure to light.

Figure 4



She then used the growth curvature bioassay to compare the IAA concentrations in the agar blocks from:

- experiment 1
- experiment 2
- experiment 3 section A
- experiment 3 section B.

Table 1 shows the scientist's results.

Table 1

Experiment	Degree of curvature in Bioassay / degrees
1	17.69
2	17.61
3A	11.22
3B	6.50

0	3	.	3
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State **two** conclusions about IAA that you can make from the results shown in **Table 1**.

[2 marks]

1 _____

2 _____

1

Turn over for the next question

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